[MP2] No Test case passes KmemCache for file does not exist

2 2

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All Sections

Hello sorry for bothering...
I've already read related questions, but my problem persists. Although my output looks mostly correct and I feel like I'm close, I'm worried I might get a 0 on MP2, but my implementation is nearly correct—it's not completely empty!

For every test case, I keep getting the same error:
"KmemCache for file does not exist."
I've been stuck on this for hours.

I'm unable to earn any points from the test cases, and it's very frustrating after working on this for more than 20 hours. :/

I would appreciate any comments or help on how I might pass at least some of the test cases, especially the simpler ones like `$ echo Ok > ok`.

Slab_checkout.out

*** Now run 'gdb' in another window.

xv6 kernel is booting

[MP2] Slab size: 4
[SLAB] New kmem_cache (name: file, object_size: 504 bytes, at: 0x0000000087f59000, max objects per slab: 8, support in cache obj: 0) is created
hart 2 starting
hart 1 starting
[SLAB] Alloc request on cache file
[SLAB] A new slab 0x0000000087f4e000 (file) is allocated
[SLAB] Object 0x0000000087f4edd0 in slab 0x0000000087f4e000 (file) is allocated and initialized
init: starting sh
[SLAB] Alloc request on cache file
[SLAB] Object 0x0000000087f4ebd8 in slab 0x0000000087f4e000 (file) is allocated and initialized
[SLAB] Free 0x0000000087f4ebd8 in slab 0x0000000087f4e000 (file)
[SLAB] end of free
$ mp2

[SLAB] kmem_cache { name: file, object_size: 504, at: 0x0000000087f59000, in_cache_obj: 0 }
[SLAB] [partial slabs]
[SLAB] [slab 0x0000000087f4e000] { freelist: 0x0000000087f4ebd8, nxt: 0xffffffffffffffff }
[SLAB] [idx 0] { addr: 0x0000000087f4e200, as_ptr: 0x0000000000000000, as_obj: {tp: 0, ref: 0, readable: 84215045, writable: 84215045, pipe: 0x0505050505050505, ip: 0x0505050505050505, off: 84215045, major: 84215045} }
[SLAB] [idx 1] { addr: 0x0000000087f4e3f8, as_ptr: 0x0000000087f4e200, as_obj: {tp: -2013994496, ref: 0, readable: 84215045, writable: 84215045, pipe: 0x0505050505050505, ip: 0x0505050505050505, off: 84215045, major: 84215045} }
[SLAB] [idx 2] { addr: 0x0000000087f4e5f0, as_ptr: 0x0000000087f4e3f8, as_obj: {tp: -2013993992, ref: 0, readable: 84215045, writable: 84215045, pipe: 0x0505050505050505, ip: 0x0505050505050505, off: 84215045, major: 84215045} }
[SLAB] [idx 3] { addr: 0x0000000087f4e7e8, as_ptr: 0x0000000087f4e5f0, as_obj: {tp: -2013993488, ref: 0, readable: 84215045, writable: 84215045, pipe: 0x0505050505050505, ip: 0x0505050505050505, off: 84215045, major: 84215045} }
[SLAB] [idx 4] { addr: 0x0000000087f4e9e0, as_ptr: 0x0000000087f4e7e8, as_obj: {tp: -2013992984, ref: 0, readable: 84215045, writable: 84215045, pipe: 0x0505050505050505, ip: 0x0505050505050505, off: 84215045, major: 84215045} }
[SLAB] [idx 5] { addr: 0x0000000087f4ebd8, as_ptr: 0x0000000087f4e9e0, as_obj: {tp: -2013992480, ref: 0, readable: 0, writable: 0, pipe: 0x0000000000000000, ip: 0x8003548800000000, off: -2147265400, major: 0} }
[SLAB] [idx 6] { addr: 0x0000000087f4edd0, as_ptr: 0x0000000900000003, as_obj: {tp: 3, ref: 9, readable: 257, writable: 0, pipe: 0x0000000000000000, ip: 0x8003548800000000, off: -2147265400, major: 0} }
[SLAB] print_kmem_cache end
$ qemu-system-riscv64: terminating on signal 15 from pid 49532 (make)

mp2-0.out
*** Now run 'gdb' in another window.

xv6 kernel is booting

[MP2] Slab size: 4
[SLAB] New kmem_cache (name: file, object_size: 504 bytes, at: 0x0000000087f59000, max objects per slab: 8, support in cache obj: 0) is created
hart 2 starting
hart 1 starting
[SLAB] Alloc request on cache file
[SLAB] A new slab 0x0000000087f4e000 (file) is allocated
[SLAB] Object 0x0000000087f4edd0 in slab 0x0000000087f4e000 (file) is allocated and initialized
init: starting sh
[SLAB] Alloc request on cache file
[SLAB] Object 0x0000000087f4ebd8 in slab 0x0000000087f4e000 (file) is allocated and

initialized
[SLAB] Free 0x0000000087f4ebd8 in slab 0x0000000087f4e000 (file)
[SLAB] end of free
$ mp2
[SLAB] kmem_cache { name: file, object_size: 504, at: 0x0000000087f59000, in_cache_obj: 0 }
[SLAB] [partial slabs]
[SLAB] [slab 0x0000000087f4e000] { freelist: 0x0000000087f4ebd8, nxt: 0xffffffffffffffff }
[SLAB] [idx 0] { addr: 0x0000000087f4e200, as_ptr: 0x0000000000000000, as_obj: {tp: 0, ref: 0, readable: 84215045, writable: 84215045, pipe: 0x0505050505050505, ip: 0x0505050505050505, off: 84215045, major: 84215045} }
[SLAB] [idx 1] { addr: 0x0000000087f4e3f8, as_ptr: 0x0000000087f4e200, as_obj: {tp: -2013994496, ref: 0, readable: 84215045, writable: 84215045, pipe: 0x0505050505050505, ip: 0x0505050505050505, off: 84215045, major: 84215045} }
[SLAB] [idx 2] { addr: 0x0000000087f4e5f0, as_ptr: 0x0000000087f4e3f8, as_obj: {tp: -2013993992, ref: 0, readable: 84215045, writable: 84215045, pipe: 0x0505050505050505, ip: 0x0505050505050505, off: 84215045, major: 84215045} }
[SLAB] [idx 3] { addr: 0x0000000087f4e7e8, as_ptr: 0x0000000087f4e5f0, as_obj: {tp: -2013993488, ref: 0, readable: 84215045, writable: 84215045, pipe: 0x0505050505050505, ip: 0x0505050505050505, off: 84215045, major: 84215045} }
[SLAB] [idx 4] { addr: 0x0000000087f4e9e0, as_ptr: 0x0000000087f4e7e8, as_obj: {tp: -2013992984, ref: 0, readable: 84215045, writable: 84215045, pipe: 0x0505050505050505, ip: 0x0505050505050505, off: 84215045, major: 84215045} }
[SLAB] [idx 5] { addr: 0x0000000087f4ebd8, as_ptr: 0x0000000087f4e9e0, as_obj: {tp: -2013992480, ref: 0, readable: 0, writable: 0, pipe: 0x0000000000000000, ip: 0x8003548800000000, off: -2147265400, major: 0} }
[SLAB] [idx 6] { addr: 0x0000000087f4edd0, as_ptr: 0x0000000900000003, as_obj: {tp: 3, ref: 9, readable: 257, writable: 0, pipe: 0x0000000000000000, ip: 0x8003548800000000, off: -2147265400, major: 0} }
[SLAB] print_kmem_cache end
$ qemu-system-riscv64: terminating on signal 15 from pid 50078 (make)

and mp2-1.out
*** Now run 'gdb' in another window.

xv6 kernel is booting

[MP2] Slab size: 4
[SLAB] New kmem_cache (name: file, object_size: 504 bytes, at: 0x0000000087f59000, max objects per slab: 8, support in cache obj: 0) is created
hart 1 starting
hart 2 starting
[SLAB] Alloc request on cache file
[SLAB] A new slab 0x0000000087f4e000 (file) is allocated
[SLAB] Object 0x0000000087f4edd0 in slab 0x0000000087f4e000 (file) is allocated and

```
initialized
init: starting sh
[SLAB] Alloc request on cache file
[SLAB] Object 0x0000000087f4ebd8 in slab 0x0000000087f4e000 (file) is allocated and
initialized
[SLAB] Free 0x0000000087f4ebd8 in slab 0x0000000087f4e000 (file)
[SLAB] end of free
$ echo Ok > ok
[SLAB] Alloc request on cache file
[SLAB] Object 0x0000000087f4ebd8 in slab 0x0000000087f4e000 (file) is allocated and
initialized
[SLAB] Free 0x0000000087f4ebd8 in slab 0x0000000087f4e000 (file)
[SLAB] end of free
$ qemu-system-riscv64: terminating on signal 15 from pid 50624 (make)
```

TESTCASE RESULT

```
== Test Slab check (Max [3(+5): slab structure (with bonus)] + [2: max objects]) (10%) ==
Slab check (Max [3(+5): slab structure (with bonus)] + [2: max objects]): FAIL (0/10) (3.0s)
   KmemCache for file does not exist
   QEMU output saved to out/slab_check.out.case
== Test public/mp2-0 (3%) ==
public/mp2-0: FAIL (0/3) (2.8s)
   KmemCache for file does not exist
   QEMU output saved to out/public/mp2-0.out.case
== Test public/mp2-1 (3%) ==
public/mp2-1: FAIL (0/3) (2.8s)
   KmemCache for file does not exist
   QEMU output saved to out/public/mp2-1.out.case
== Test public/mp2-2 (3%) ==
public/mp2-2: FAIL (0/3) (2.7s)
   KmemCache for file does not exist
   QEMU output saved to out/public/mp2-2.out.case
== Test public/mp2-3 (3%) ==
Timeout! public/mp2-3: FAIL (0/3) (21.9s)
   KmemCache for file does not exist
   QEMU output saved to out/public/mp2-3.out.case
== Test public/mp2-4 (3%) ==.
... same for all or timeout
```

thanks anyways
Mehmet Efe Sak

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MEHMET EFE SAK (MEHMET EFE SAK) (https://cool.ntu.edu.tw/courses/46296/users/244796)
Monday

```
struct kmem_cache *kmem_cache_create(char *name, uint object_size) {
  if (object_size == 0 || object_size > PGSIZE) {
    object_size = sizeof(void *);
  }

  struct kmem_cache *cache = (struct kmem_cache *)kalloc();
  if (!cache) {
    panic("[SLAB] kmem_cache_create: Failed to allocate cache structure");
  }

  memset(cache, 0, sizeof(*cache));

  strncpy(cache->name, name ? name : "default", sizeof(cache->name) - 1);
  cache->name[sizeof(cache->name) - 1] = '\0';

  cache->object_size = ALIGN_UP(object_size, sizeof(void *));
  initlock(&cache->lock, cache->name);
  cache->partial = 0;
  cache->free = 0;

  // Calculate the number of in-cache objects
  uint in_cache_space = PGSIZE - sizeof(struct kmem_cache);
  cache->in_cache_objs = in_cache_space / cache->object_size;
  cache->in_cache_in_use = 0;

  // Allocate space for in-cache objects
  if (cache->in_cache_objs > 0) {
    cache->in_cache_objects = (void *)((char *)cache + sizeof(struct kmem_cache));
  } else {
```

```
    cache->in_cache_objects = 0;
  }

  debug("[SLAB] New kmem_cache (name: %s, object_size: %d bytes, at: %p, max objects
per slab: %d, in-cache objs: %d) is created\n",
        cache->name, cache->object_size, cache, MAX_OBJECTS_PER_SLAB, cache-
>in_cache_objs);

  return cache;
}
```

I think it creates knem cash because outputs have line [SLAB] New kmem_cache (name: file, object_size: 504 bytes, at: 0x0000000087f59000, max objects per slab: 8, in-cache objs: 7) is created

but still got the eror KmemCache for file does not exist

Reply



詹文淵 (CHAN,WEN-YUAN) (https://cool.ntu.edu.tw/courses/46296/users/144411)

Monday

In line "[SLAB] New kmem_cache (name: file, object_size: 504 bytes, ...", "object_size" should be "object size".

Reply